In [ ]:
```
# Name : Richa Patel
# Course Code and Name: 3CS502ME24 OPEN SOURCE TECHNOLOGIES
# Roll no : 24BTM058
# Practical NO : 8
# Aim : Develop a Jupyter notebook to acquire foundational skills in data manipu
# Methodology Followed:
```

In [1]:
```
import pandas as pd
import numpy as np
```

In [2]:
```
df = pd.read_csv("BostonHousing.csv")
df
```

Out[2]:

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	b	l
0	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296	15.3	396.90	4
1	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8	396.90	9
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8	392.83	4
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	394.63	2
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	396.90	5
...	...	...	...	...	...	...	...	...	...	...	...	...	
501	0.06263	0.0	11.93	0	0.573	6.593	69.1	2.4786	1	273	21.0	391.99	9
502	0.04527	0.0	11.93	0	0.573	6.120	76.7	2.2875	1	273	21.0	396.90	9
503	0.06076	0.0	11.93	0	0.573	6.976	91.0	2.1675	1	273	21.0	396.90	5
504	0.10959	0.0	11.93	0	0.573	6.794	89.3	2.3889	1	273	21.0	393.45	6
505	0.04741	0.0	11.93	0	0.573	6.030	80.8	2.5050	1	273	21.0	396.90	7

506 rows × 14 columns

In [3]:
```
df.head()
df.tail()
```

Out[3]:

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	b	lst
501	0.06263	0.0	11.93	0	0.573	6.593	69.1	2.4786	1	273	21.0	391.99	9.
502	0.04527	0.0	11.93	0	0.573	6.120	76.7	2.2875	1	273	21.0	396.90	9.
503	0.06076	0.0	11.93	0	0.573	6.976	91.0	2.1675	1	273	21.0	396.90	5.
504	0.10959	0.0	11.93	0	0.573	6.794	89.3	2.3889	1	273	21.0	393.45	6.
505	0.04741	0.0	11.93	0	0.573	6.030	80.8	2.5050	1	273	21.0	396.90	7.

In [4]:
```
df.shape
df.columns
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 506 entries, 0 to 505
Data columns (total 14 columns):
 #   Column   Non-Null Count  Dtype  
---  ------   --------------  -----  
 0   crim     506 non-null    float64
 1   zn       506 non-null    float64
 2   indus    506 non-null    float64
 3   chas     506 non-null    int64  
 4   nox      506 non-null    float64
 5   rm       506 non-null    float64
 6   age      506 non-null    float64
 7   dis      506 non-null    float64
 8   rad      506 non-null    int64  
 9   tax      506 non-null    int64  
 10  ptratio  506 non-null    float64
 11  b        506 non-null    float64
 12  lstat    506 non-null    float64
 13  medv     506 non-null    float64
dtypes: float64(11), int64(3)
memory usage: 55.5 KB
```

In [5]:
```
df.describe()
```

Out[5]:

	crim	zn	indus	chas	nox	rm	a
count	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.0000
mean	3.613524	11.363636	11.136779	0.069170	0.554695	6.284634	68.5749
std	8.601545	23.322453	6.860353	0.253994	0.115878	0.702617	28.1488
min	0.006320	0.000000	0.460000	0.000000	0.385000	3.561000	2.9000
25%	0.082045	0.000000	5.190000	0.000000	0.449000	5.885500	45.0250
50%	0.256510	0.000000	9.690000	0.000000	0.538000	6.208500	77.5000
75%	3.677083	12.500000	18.100000	0.000000	0.624000	6.623500	94.0750
max	88.976200	100.000000	27.740000	1.000000	0.871000	8.780000	100.0000

In [6]:
```
df.isnull().sum()
```

Out[6]:
```
crim       0
zn         0
indus      0
chas       0
nox        0
rm         0
age        0
dis        0
rad        0
tax        0
ptratio    0
b          0
lstat      0
medv       0
dtype: int64
```

In [8]:
```
df.isna().any()
```

Out[8]:
```
crim       False
zn         False
indus      False
chas       False
nox        False
rm         False
age        False
dis        False
rad        False
tax        False
ptratio    False
b          False
lstat      False
medv       False
dtype: bool
```

In [9]:
```
df[["rm", "lstat", "medv"]]
```

Out[9]:

	rm	lstat	medv
0	6.575	4.98	24.0
1	6.421	9.14	21.6
2	7.185	4.03	34.7
3	6.998	2.94	33.4
4	7.147	5.33	36.2
...	...	...	...
501	6.593	9.67	22.4
502	6.120	9.08	20.6
503	6.976	5.64	23.9
504	6.794	6.48	22.0
505	6.030	7.88	11.9

506 rows × 3 columns

In [10]:
```
df[df["medv"] > 30]
```

Out[10]:

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	b
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8	392.83
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	394.63
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	396.90
39	0.02763	75.0	2.95	0	0.428	6.595	21.8	5.4011	3	252	18.3	395.63
40	0.03359	75.0	2.95	0	0.428	7.024	15.8	5.4011	3	252	18.3	395.62
...	...	...	...	...	...	...	...	...	...	...	...	...
368	4.89822	0.0	18.10	0	0.631	4.970	100.0	1.3325	24	666	20.2	375.52
369	5.66998	0.0	18.10	1	0.631	6.683	96.8	1.3567	24	666	20.2	375.33
370	6.53876	0.0	18.10	1	0.631	7.016	97.5	1.2024	24	666	20.2	392.05
371	9.23230	0.0	18.10	0	0.631	6.216	100.0	1.1691	24	666	20.2	366.15
372	8.26725	0.0	18.10	1	0.668	5.875	89.6	1.1296	24	666	20.2	347.88

84 rows × 14 columns

In [11]:
```
df[(df["rm"] > 6) & (df["medv"] > 25)]
```

Out[11]:

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	b
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8	392.83
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	394.63
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	396.90
5	0.02985	0.0	2.18	0	0.458	6.430	58.7	6.0622	3	222	18.7	394.12
7	0.14455	12.5	7.87	0	0.524	6.172	96.1	5.9505	5	311	15.2	396.90
...	...	...	...	...	...	...	...	...	...	...	...	...
369	5.66998	0.0	18.10	1	0.631	6.683	96.8	1.3567	24	666	20.2	375.33
370	6.53876	0.0	18.10	1	0.631	7.016	97.5	1.2024	24	666	20.2	392.05
371	9.23230	0.0	18.10	0	0.631	6.216	100.0	1.1691	24	666	20.2	366.15
409	14.43830	0.0	18.10	0	0.597	6.852	100.0	1.4655	24	666	20.2	179.36
473	4.64689	0.0	18.10	0	0.614	6.980	67.6	2.5329	24	666	20.2	374.68

119 rows × 14 columns

In [12]:
```
df.sort_values("medv", ascending=False)
```

Out[12]:

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	b
369	5.66998	0.0	18.1	1	0.631	6.683	96.8	1.3567	24	666	20.2	375.33
372	8.26725	0.0	18.1	1	0.668	5.875	89.6	1.1296	24	666	20.2	347.88
371	9.23230	0.0	18.1	0	0.631	6.216	100.0	1.1691	24	666	20.2	366.15
370	6.53876	0.0	18.1	1	0.631	7.016	97.5	1.2024	24	666	20.2	392.05
368	4.89822	0.0	18.1	0	0.631	4.970	100.0	1.3325	24	666	20.2	375.52
...	...	...	...	...	...	...	...	...	...	...	...	...
414	45.74610	0.0	18.1	0	0.693	4.519	100.0	1.6582	24	666	20.2	88.27
399	9.91655	0.0	18.1	0	0.693	5.852	77.8	1.5004	24	666	20.2	338.16
400	25.04610	0.0	18.1	0	0.693	5.987	100.0	1.5888	24	666	20.2	396.90
398	38.35180	0.0	18.1	0	0.693	5.453	100.0	1.4896	24	666	20.2	396.90
405	67.92080	0.0	18.1	0	0.693	5.683	100.0	1.4254	24	666	20.2	384.97

506 rows × 14 columns

In [13]:
```
df.nlargest(10, "medv")
```

Out[13]:

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	b
161	1.46336	0.0	19.58	0	0.6050	7.489	90.8	1.9709	5	403	14.7	374.43
162	1.83377	0.0	19.58	1	0.6050	7.802	98.2	2.0407	5	403	14.7	389.61
163	1.51902	0.0	19.58	1	0.6050	8.375	93.9	2.1620	5	403	14.7	388.45
166	2.01019	0.0	19.58	0	0.6050	7.929	96.2	2.0459	5	403	14.7	369.30
186	0.05602	0.0	2.46	0	0.4880	7.831	53.6	3.1992	3	193	17.8	392.63
195	0.01381	80.0	0.46	0	0.4220	7.875	32.0	5.6484	4	255	14.4	394.23
204	0.02009	95.0	2.68	0	0.4161	8.034	31.9	5.1180	4	224	14.7	390.55
225	0.52693	0.0	6.20	0	0.5040	8.725	83.0	2.8944	8	307	17.4	382.00
257	0.61154	20.0	3.97	0	0.6470	8.704	86.9	1.8010	5	264	13.0	389.70
267	0.57834	20.0	3.97	0	0.5750	8.297	67.0	2.4216	5	264	13.0	384.54

In [15]:
```
df["chas"].unique()
df["chas"].value_counts()
df["rad"].value_counts()
```

Out[15]:
```
rad
24    132
5     115
4     110
3      38
6      26
8      24
2      24
1      20
7      17
Name: count, dtype: int64
```

In [16]:
```
print(df["medv"].mean())
print(df["medv"].median())
print(df["medv"].max())
print(df["medv"].min())
```

```
22.532806324110677
21.2
50.0
5.0
```

In [17]:
```
df.groupby("rad")["medv"].agg(["mean", "min", "max", "count"])
```

Out[17]:

rad	mean	min	max	count
1	24.365000	11.9	50.0	20
2	26.833333	15.7	43.8	24
3	27.928947	14.4	50.0	38
4	21.387273	7.0	50.0	110
5	25.706957	11.8	50.0	115
6	20.976923	16.8	24.8	26
7	27.105882	17.6	42.8	17
8	30.358333	16.0	50.0	24
24	16.403788	5.0	50.0	132

In [18]:
```
df.agg({
    "crim": ["mean", "max"],
    "rm": ["mean", "max"],
    "medv": ["mean", "min", "max"]
})
```

Out[18]:

	crim	rm	medv
mean	3.613524	6.284634	22.532806
max	88.976200	8.780000	50.000000
min	NaN	NaN	5.000000

In [19]:
```
df["price_category"] = pd.cut(
    df["medv"],
    bins=[0, 20, 30, 40, 60],
    labels=["Low", "Medium", "High", "Very High"]
)

df[["medv", "price_category"]].head()
```

Out[19]:

	medv	price_category
0	24.0	Medium
1	21.6	Medium
2	34.7	High
3	33.4	High
4	36.2	High

In [20]:
```
df["price_category"].value_counts()
```

Out[20]:
```
price_category
Low          215
Medium       207
High          53
Very High     31
Name: count, dtype: int64
```

In [21]:
```
df = df.rename(columns={"medv": "MEDV"})
df.head()
```

Out[21]:

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	b	lsta
0	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296	15.3	396.90	4.9
1	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8	396.90	9.1
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8	392.83	4.0
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	394.63	2.9
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	396.90	5.3

In [ ]: